saying that it *is* real. The point is that quantum calculations unambiguously tell us the probability that an electron will land at one or another point on the screen, and these predictions agree with the data, spot on. As far as the theory's verification and predictive utility are concerned, the story we tell of how the electron got to that point on the screen is of little relevance.

But surely, you'd continue to press, we can settle the issue of what really happens by changing the experimental setup so that we can also watch the supposed fuzzy mélange of possible pasts melding into the observed present. It's a good suggestion, but we already know that there has to be a hitch. In Chapter 4, we learned that probability waves are not directly observable; since Feynman's coalescing histories are nothing but a particular way of thinking about probability waves, they, too, must evade direct observation. And they do. Observations cannot tease apart individual histories; rather, observations reflect *averages* of all possible histories. So, if you change the setup to observe the electrons in flight, you will see each electron pass by your additional detector in one location or another; you will never see any fuzzy multiple histories. When you use quantum mechanics to explain *why* you saw the electron in one place or another, the answer will involve averaging over all possible histories that could have led to that intermediate observation. But the observation itself has access only to histories that have already merged. By looking at the electron in flight, you have merely pushed back the notion of what you mean by a history. Quantum mechanics is starkly efficient: it explains what you see but prevents you from seeing the explanation.

You might further ask: Why, then, is classical physics—commonsense physics—which describes motion in terms of unique histories and trajectories, at all relevant to the universe? Why does it work so well in explaining and predicting the motion of everything from baseballs to planets to comets? How come there is no evidence in day-to-day life of the strange way in which the past apparently unfolds into the present? The reason, discussed briefly in Chapter 4 and to be elaborated shortly with greater precision, is that baseballs, planets, and comets are comparatively large, at least when compared with particles like electrons. And in quantum mechanics, the larger something is, the more skewed the averaging becomes: All possible trajectories *do* contribute to